6 A single-turn copper square frame of length L is rotated with constant angular speed ω by an external torque in a constant magnetic field of flux density B. The frame rotates counter-clockwise about the axis of rotation as shown in Fig. 6.1.

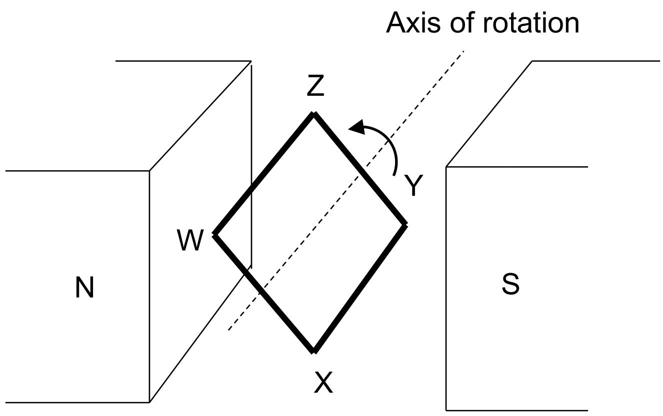

Fig. 6.1

Fig. 6.2 shows the side view of the coil when WX is at an angle θ above the horizontal.

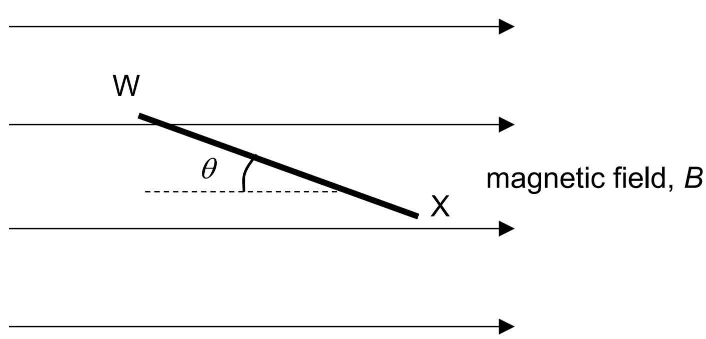

Fig. 6.2

(a) State the direction of the induced current in the frame.

In the direction WXYZW. [1] (accept WX)

.... [1]

(b) Explain why an external torque is required to maintain the rotation of the frame at a constant angular speed.

The changing magnetic flux linkage produces an induced emf, and hence current due to Faraday's law. [1]

The force due to this induced current opposes the motion by Lenz' law. [1]

.... [2]

(c) **(i)** At the instant shown in Fig. 6.2, write down an expression for the flux linkage in the coil in terms of B, θ and L.

The flux through the frame is $\phi = (B\sin\theta)L^2$

[1]

(ii) Hence show that the magnitude of the induced e.m.f. in the coil at this instant is $|\varepsilon| = BL^2\omega\cos\theta$.

$\phi = BL^2 \sin\omega t$ [1]

By Faraday's Law,

$$|\varepsilon| = \left|\frac{d\phi}{dt}\right| = BL^2\omega\cos\omega t = BL^2\omega\cos\theta \quad [1]$$

[2]

(d) The resistance of the frame is 5.0 Ω and the length L of the square frame is 0.20 m. The frame is rotated in the magnetic field of flux density 1.0 T at an angular frequency of 10 rad s^{-1}. Using your expression in **(c)(ii)**, calculate the average power dissipated in the frame.

$$P = \frac{\varepsilon^2}{R}$$

$$P = \frac{B^2L^4\omega^2\cos^2\theta}{R} = \frac{B^2L^4\omega^2\cos^2\omega t}{R} = P_0\cos^2\omega t \quad [1]$$

$$<P> = \frac{P_0}{2} \quad [1]$$

$$<P> = \frac{B^2L^4\omega^2}{2R} = \frac{1.0^2 \times 0.20^4 \times 10^2}{2\times 5.0} = 16 \text{ mW} \quad [1]$$

average power dissipated = ... W [3]

[Total: 9]

7 (a) (i) Describe the *photoelectric effect* in terms of energy.